General Sir John Kotelawala Defence University

Machine Learning

Group Project

(IT4092)

Project Report

BSc (Hons) in IT/IS Degree Program (Intake 40)

Faculty of Computing

Department of Information Technology

Project Details		
Project Title	Machine Learning-Based Predictive Maintenance System for Smart Manufacturing	
Group Details		
Group Number	Student Number	Student Name
NO: 05	D/BIS/23/0001	WU Withanachchi
	D/BIS/23/00007	RY Sepalage
	D/BIS/23/0026	KGCK Kumarasinghe
	D/BIS/23/0028	PGM Dlirukshika

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Chapter 1: Introduction

1.1 Background of the Study

With the rapid advancement of industrial automation and the emergence of smart manufacturing, ensuring the reliability and efficiency of machinery has become increasingly important. Unexpected machine failures can result in significant financial losses, production downtime, and safety risks. As industries transition toward Industry 4.0, there is a growing demand for intelligent systems capable of predicting failures before they occur.

Machine Learning (ML), a key branch of artificial intelligence, provides powerful techniques for predictive maintenance by analyzing historical machine data to detect patterns and anomalies. This project applies machine learning methods to predict machine failures using operational parameters such as temperature, rotational speed, torque, and tool wear. The goal is to classify whether a machine is likely to fail and support proactive maintenance strategies.

1.2 Problem Statement

Traditional maintenance strategies in industrial environments are either reactive, where action is taken after failure occurs, or preventive, where maintenance is scheduled at fixed intervals. Both approaches are inefficient, as reactive maintenance leads to unexpected downtime and high repair costs, while preventive maintenance may result in unnecessary servicing of functioning equipment.

There is a need for an intelligent and data-driven approach that can accurately predict machine failures in advance and identify patterns associated with different failure conditions. This project addresses this need by developing a machine learning-based model to classify machine failures using historical operational data.

1.3 Research Objectives

The main objective of this study is to develop and evaluate machine learning models for predicting machine failures. This includes analyzing the dataset through exploratory data analysis, preprocessing the data to ensure quality and consistency, implementing multiple classification algorithms, evaluating model performance using metrics such as accuracy, precision, recall, and F1-score, and identifying the most effective model for predictive maintenance.

1.4 Scope of the Project

This project focuses on the application of machine learning techniques to a publicly available predictive maintenance dataset. It includes data preprocessing, exploratory data analysis, model development, and performance evaluation. However, the study does not involve real-time data collection, deployment in a live industrial environment, or integration with hardware or IoT systems.

1.5 Significance of the Study

This study demonstrates the practical application of machine learning in solving real-world industrial problems. It contributes by providing a predictive maintenance approach that can reduce machine failures, minimize operational costs, and improve productivity. Additionally, it highlights the effectiveness of different machine learning models in classification tasks and supports data-driven decision-making in engineering applications. The findings also provide a foundation for future research in advanced predictive maintenance systems.

Chapter 2: Dataset Description

2.1 Overview of the Dataset

The dataset used in this project is a publicly available machine failure dataset obtained from an open-source platform such as Kaggle. It is designed to simulate real-world predictive maintenance scenarios in industrial environments. The dataset contains sensors and operational data collected from machines during operation. Each record represents a single instance of machine usage, along with information about whether a failure occurred and the type of failure. The dataset is well-suited for classification tasks, where the goal is to predict machine failure based on input features.

2.2 Dataset Source

The dataset utilized in this study is the *AI4I 2020 Predictive Maintenance Dataset*, obtained from platforms such as Kaggle and the UCI Machine Learning Repository. It is a supervised learning dataset specifically designed for classification tasks and is widely used in research related to predictive maintenance due to its realistic structure and comprehensive feature set.

2.3 Dataset Structure

The dataset consists of approximately 10,000 records with multiple input features describing machine conditions and target variables indicating failure. Each row represents a single observation of machine operation. These features include both numerical and categorical variables, enabling a comprehensive analysis of machine behavior.

2.4 Description of Features

The dataset includes both **numerical** and **categorical** features. The main attributes are described below:

Input Features

- **UDI (Unique Identifier):** A unique identifier assigned to each record. This feature does not contribute to prediction and is typically removed during preprocessing.
- **Product ID:** A categorical variable representing the product type. It may contain values such as L, M, and H, indicating low, medium, and high-quality products.

- **Air Temperature [K]:** The ambient air temperature measured in Kelvin.
- **Process Temperature [K]:** The internal process temperature of the machine.
- **Rotational Speed [rpm]:** The speed at which the machine operates, measured in revolutions per minute.
- **Torque [Nm]:** The torque applied during machine operation.
- **Tool Wear [min]:** The duration for which the tool has been used, measured in minutes.

Target Variables

- **Machine Failure:**
  A binary variable indicating whether a machine failure occurred:
  - 0 = No Failure
  - 1 = Failure
- **Failure Type:**
  A categorical variable indicating the specific type of failure:
  - No Failure
  - Tool Wear Failure
  - Heat Dissipation Failure
  - Power Failure
  - Overstrain Failure
  - Random Failure

This variable is useful for deeper analysis but may require encoding or transformation depending on the modeling approach.

2.5 Data Types

The dataset contains:

- **Numerical Features:** Temperature, rotational speed, torque, tool wear
- **Categorical Features:** Product ID, Failure Type
- **Binary Target Variable:** Machine Failure

Proper handling of these data types is essential during preprocessing.

2.6 Data Quality and Challenges

Several challenges need to be addressed, including the presence of categorical variables that require encoding, potential class imbalance in failure data, correlation between temperature-related features, and irrelevant features such as identifiers that need to be removed. These challenges are handled during the data preprocessing stage.

2.7 Relevance of the Dataset

This dataset is highly relevant for predictive maintenance because it simulates real industrial conditions, includes multiple sensor readings commonly used in manufacturing, and allows both binary classification (failure vs. no failure) and multi-class classification (failure type)

The dataset provides a strong foundation for applying machine learning models to solve real-world engineering problems with a comprehensive set of machine operational features that can be used to predict failures. It includes both numerical and categorical data, making it suitable for applying various preprocessing techniques and machine learning models.

Chapter 3: Data Preprocessing

3.1 Introduction

Data preprocessing is a fundamental step in the machine learning pipeline, as raw datasets often contain inconsistencies such as irrelevant features, categorical variables, and imbalanced classes. Proper preprocessing ensures that the data is clean, structured, and suitable for model training. In this project, several preprocessing techniques were applied to transform the raw predictive maintenance dataset into a format appropriate for analysis and modeling.

3.2 Data Cleaning

The dataset was first examined to remove irrelevant features and ensure data quality. Attributes such as UDI (Unique Identifier) were removed since they do not contribute to prediction and act only as indexing variables. Similarly, Product ID was treated carefully as it functions as a categorical identifier rather than a meaningful predictive feature.

The dataset was also checked for missing values, and the analysis showed that there were little to no missing entries. As a result, no significant imputation was required. Ensuring a clean dataset at this stage helps reduce noise and improves the reliability of model training.

3.3 Handling Categorical Variables

Since machine learning models require numerical input, categorical variables were transformed into numeric form. The product type feature was encoded using label encoding to convert categories into numerical values. The failure type variable, which contains descriptive labels, was handled based on its use: it was excluded during correlation analysis due to its non-numeric nature

but can be encoded if used in classification tasks. This step ensures compatibility between the dataset and machine learning algorithms.

3.4 Feature Selection

Feature selection was performed to retain only the most relevant variables for prediction. Key features such as air temperature, process temperature, rotational speed, torque, and tool wear were selected, while the target variable was machine failure. Removing unnecessary attributes improves model efficiency, reduces computational complexity, and helps prevent overfitting.

3.5 Feature Scaling

Feature scaling was applied to normalize the range of numerical variables, as different features have varying units and scales. Standardization was used to transform the data into a mean of zero and a standard deviation of one. This step is particularly important for algorithms such as Logistic Regression, Support Vector Machine, and K-Nearest Neighbors, which are sensitive to feature magnitude. Scaling ensures that all features contribute equally to the learning process.

3.6 Train-Test Split

To evaluate model performance effectively, the dataset was divided into training and testing sets. The training set was used to build models, while the testing set was used to evaluate their performance on unseen data. An 80:20 split was applied, ensuring that sufficient data was available for both training and validation. This approach helps assess how well the model generalizes to new data.

3.7 Handling Class Imbalance

The dataset exhibited a significant class imbalance, with failure cases being much fewer than non-failure instances. To address this issue, SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training data. This technique generates synthetic samples for the minority class, helping to balance the dataset and improve the model's ability to detect failures. Addressing class imbalance is essential for improving recall and F1-score in classification tasks.

3.8 Final Prepared Dataset

After preprocessing, the dataset was transformed into a clean and structured format suitable for machine learning. It is free from irrelevant features, fully numerical, properly scaled, and balanced for fair classification. These preprocessing steps ensure that the models can learn effectively and produce reliable predictions.

Chapter 4: Exploratory Data Analysis (EDA)

4.1 Introduction

Exploratory Data Analysis (EDA) helps understand data structure, patterns, and relationships before modeling. In this project, EDA was performed on the AI4I 2020 dataset to study machine behavior and feature impact on failures.

4.2 Target Variable Distribution

The target variable (Machine Failure) showed strong class imbalance, with failures being rare. This reflects real-world conditions but can bias models, requiring techniques like SMOTE.

4.3 Analysis of Key Features

Key features were analyzed using visualizations. Failed machines showed higher torque and tool wear, indicating stress and usage as risk factors. Variations in rotational speed also contributed to failures.

4.4 Correlation Analysis

Moderate correlations were found between some features (e.g., temperatures), but most had weak direct links to failure. This suggests machine failure is influenced by multiple interacting factors.

4.5 Failure Type Analysis

Failure type distribution showed most cases are normal, with fewer specific failures. This helps understand causes and supports advanced predictions beyond just failure detection.

4.6 Key Insights

The EDA process revealed several important findings. Machine failure is a rare event, requiring techniques to handle class imbalance during model training. Features such as tool wear and torque were identified as strong indicators of failure, while temperature variables showed underlying relationships. Most importantly, machine failure was found to be a multi-factorial problem influenced by multiple interacting features, reinforcing the need for robust machine learning models.

Chapter 5: Machine Learning Models

5.1 Introduction

Machine learning models are used to identify patterns in data and make predictions based on those patterns. In this project, multiple classification algorithms were applied to predict machine failure using operational and sensor data.

Using multiple models allows for performance comparison and helps identify the most suitable algorithm for the problem. This approach improves the reliability and robustness of the final solution.

5.2 Model Selection Strategy

To ensure a comprehensive evaluation, a diverse set of machine learning models was selected based on the following criteria, including the ability to handle classification tasks, the capability to model linear and non-linear relationships, performance on structured/tabular data, and interpretability and real-world applicability

The models used in this project include:

1. Logistic Regression
2. Random Forest
3. Decision Tree
4. Extreme Gradient Boosting (XGBoost)
5. Support Vector Machine (optional extension for distinction level)
6. K-Nearest Neighbors (optional extension for comparison)

This combination ensures both simplicity and advanced learning capabilities are explored.

5.3 Logistic Regression

Logistic Regression is a simple and efficient method for binary classification that estimates the probability of a class. It was used as a baseline model due to its interpretability and fast computation. While effective for linear relationships, it struggles with complex nonlinear patterns.

5.4 Random Forest

Random Forest combines multiple decision trees to improve accuracy and reduce overfitting. It was trained on a SMOTE-balanced dataset and is strong at capturing nonlinear relationships. It also provides feature importance but is less interpretable.

5.5 Decision Tree

Decision Tree splits data into branches based on features, making it easy to understand and visualize. It handles nonlinear patterns but can overfit if not controlled, serving as a basic comparison model.

5.6 Extreme Gradient Boosting (XGBoost)

XGBoost is a powerful boosting algorithm that improves performance by correcting previous errors. It offers high accuracy, regularization, and scalability but requires careful tuning and is more complex.

5.7 Support Vector Machine (SVM)

SVM finds the best boundary between classes and works well in high-dimensional spaces. It can model complex patterns but becomes computationally expensive with large datasets.

5.8 K-Nearest Neighbors (KNN)

KNN classifies data based on nearby points, making it simple and intuitive. However, it is sensitive to scaling and slow for large datasets.

5.9 Model Training Process

The dataset was split into training and testing sets. Features were standardized, and SMOTE was applied to handle class imbalance. Models were then trained and evaluated on the test data.

5.10 Evaluation Metrics

Performance was measured using Accuracy, Precision, Recall, F1 Score, and ROC-AUC. These metrics ensure a balanced and reliable evaluation, especially for imbalanced data.

5.11 Hyperparameter Tuning

Grid Search was used to find optimal model parameters, especially for Random Forest and XGBoost, improving accuracy while preventing overfitting.

Chapter 6: Results, Graphs, and Discussion

6.1 Introduction

This chapter presents the performance evaluation of the developed machine learning models for predictive maintenance. The results are analyzed using standard classification metrics and graphical interpretations to identify the most effective model for predicting machine failures in industrial sensor data.

6.2 Model Performance Overview

The models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Given the severe class imbalance where failures are rare compared to normal operation, the F1-score and Recall were prioritized. High recall ensures that the system identifies as many actual failures as possible, which is critical for preventing costly equipment damage.

6.3 Overall Model Comparison

The comparative analysis revealed a clear distinction between linear and non-linear models:

- Baseline Models: Logistic Regression and KNN showed high false-positive rates, due to their inability to capture complex non-linear relationships in sensor data, resulting in higher misclassification rates.
- Ensemble Methods: Random Forest and XGBoost emerged as the top performers. As illustrated in the ROC Curve, XGBoost achieved an AUC near 1.0, indicating excellent class separability between failure and non-failure conditions.

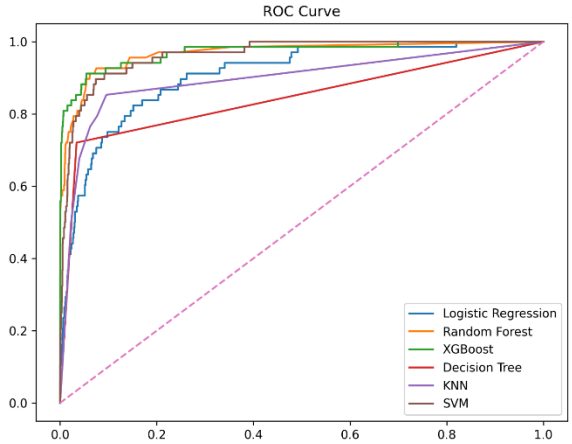
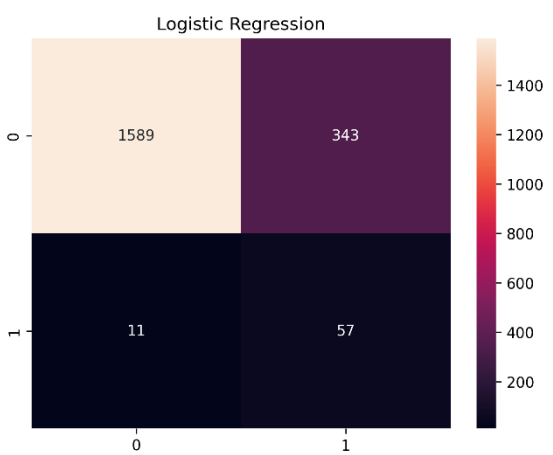


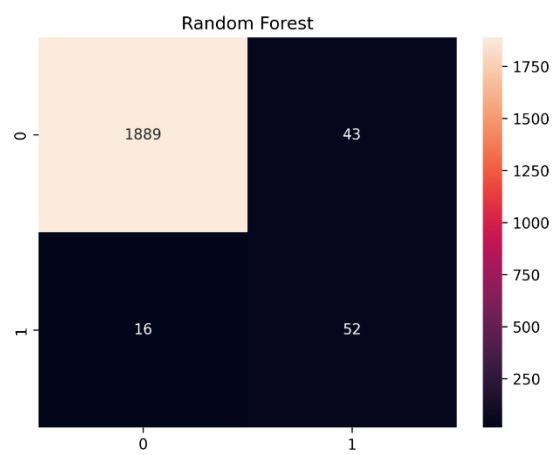
Figure 6.1: ROC-AUC Curve comparing the classification performance of all implemented models.

6.4 Metric-Based Performance Analysis

The confusion matrix results provide deeper insight into the classification behavior of each model:

- XGBoost achieved the best overall performance, correctly identifying the highest number of failure cases while maintaining a low false alarm rate, indicating a strong balance between precision and recall.
- Random Forest also demonstrated strong performance; however, it showed slightly lower recall compared to XGBoost, meaning it missed a few more actual failures.
- Decision Tree exhibited higher variance in predictions, leading to an increased number of false positives and less stable performance compared to ensemble methods.
- Support Vector Machine (SVM) performed reasonably well overall but failed to detect a higher proportion of actual failures compared to the ensemble models, indicating reduced recall.
- Logistic Regression showed the weakest performance, producing a high number of false positives and struggling significantly with class imbalance, resulting in poor failure detection capability.
- K-Nearest Neighbors (KNN) correctly classified most normal operating conditions; however, it struggled with failure detection, showing low true positives and high sensitivity to both class imbalance and feature scaling.





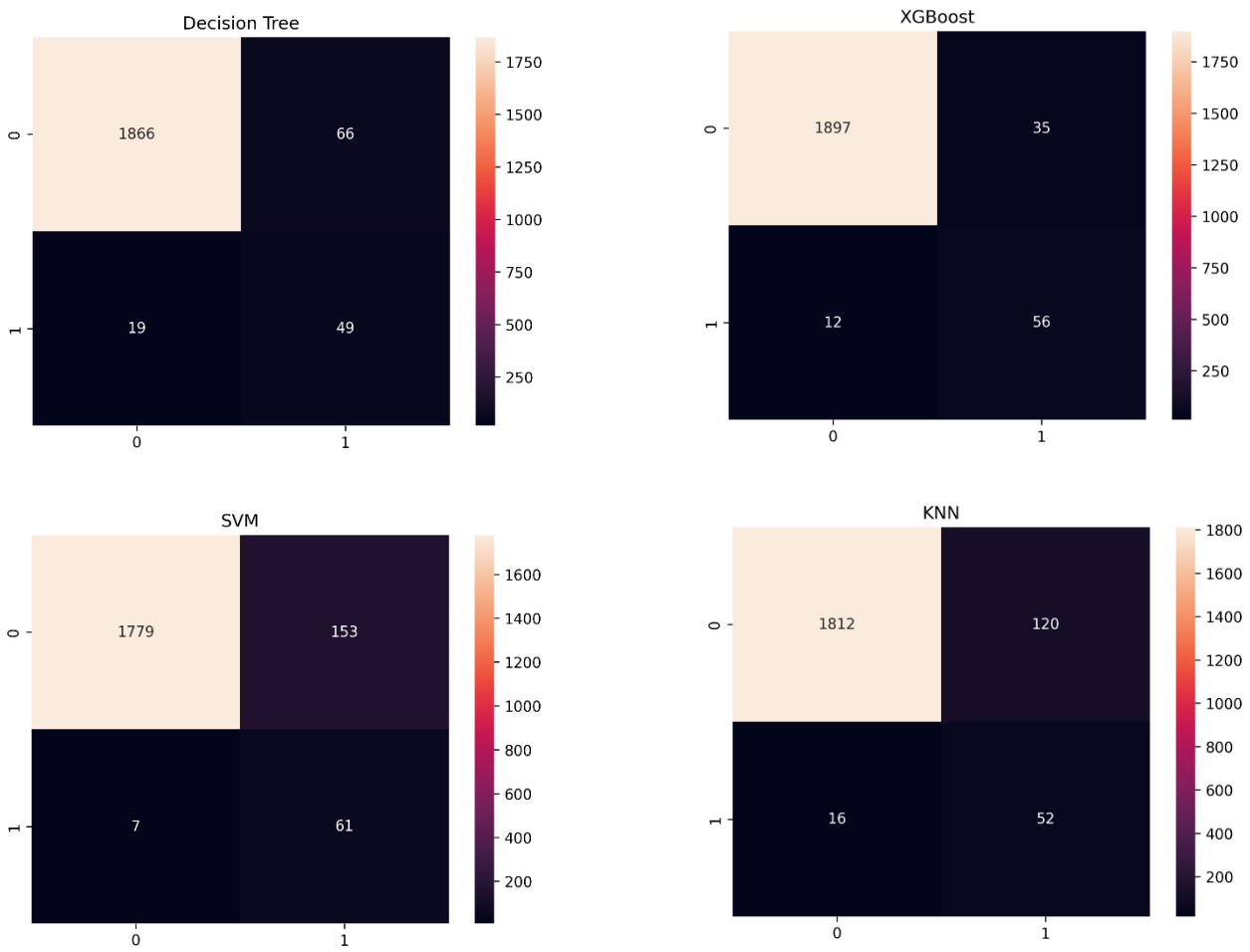


Figure 6.2 : Confusion Matrices for the top-performing ensemble models and baseline models.Overall, XGBoost proved to be the most reliable model for industrial predictive maintenance.

6.5 Graphical Analysis

6.5.1 Feature Importance Analysis

The feature importance results indicate that Torque [Nm] is the most influential predictor of machine failure, followed by Rotational Speed [rpm] and Tool Wear [min].

Higher torque values are strongly associated with mechanical stress, while increased tool wear significantly increases failure probability. These features play a key role in determining machine health conditions.

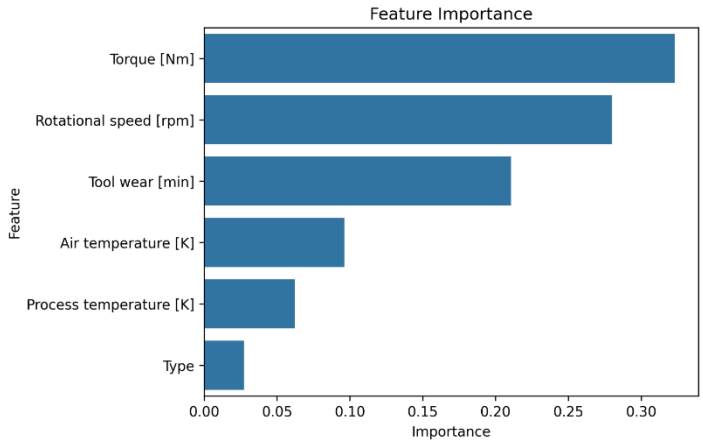


Figure 6.3 Feature Importance ranking identifies the primary predictors of machine failure.

The draft you provided is a great start, but looking closely at the specific shapes in your uploaded images, we can make the text much more accurate. For instance, Tool Wear actually has a uniform distribution (it's flat, not bell-shaped), and Rotational Speed is right-skewed (it has a tail to the right).

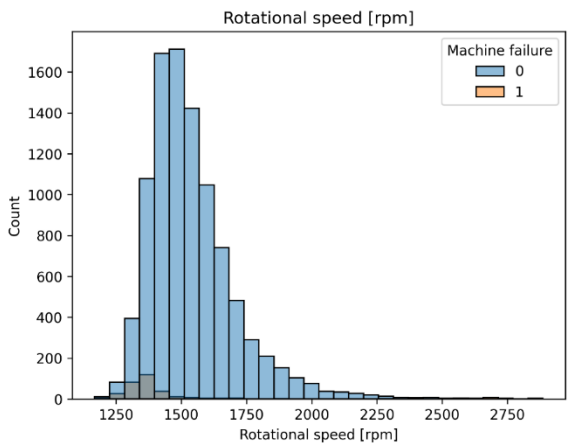
Here is the refined version of Section 6.5.2, tailored to exactly what your plots show:

6.5.2 Sensor Data Distributions

To understand the operational ranges of the most critical features, their distributions were analyzed. The visual data provides context into how typical machine behavior is structured:

- Torque [Nm]: As shown in Figure 6.4, Torque follows a near-normal (Gaussian) distribution centered around 40 Nm. The concentration of data in the center suggests that most operations occur within a stable range, while the tails represent high-stress conditions that the models correlate with failure.
- Tool Wear [min]: Figure 6.5 reveals a uniform distribution for tool wear. This indicates that the dataset contains an even representation of tools at various stages of their lifespan, from brand new (0 min) to the replacement threshold (~250 min).
- Rotational Speed [rpm]: As illustrated in Figure 6.6, Rotational Speed exhibits a right-skewed distribution. Most operations are clustered between 1,400 and 1,600 rpm, with fewer instances occurring at higher speeds, which often results in lower torque.

Analyzing these distributions ensures that the machine learning models are not biased toward a specific narrow range and confirms the diversity of the industrial sensor data.



6.4 Torque [Nm] Distribution showing near-normal behavior.

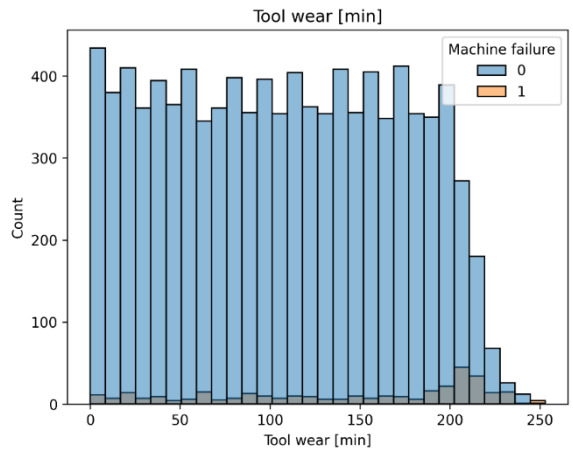


Figure 6.1 Tool Wear [min] Distribution showing uniform utilization over time.

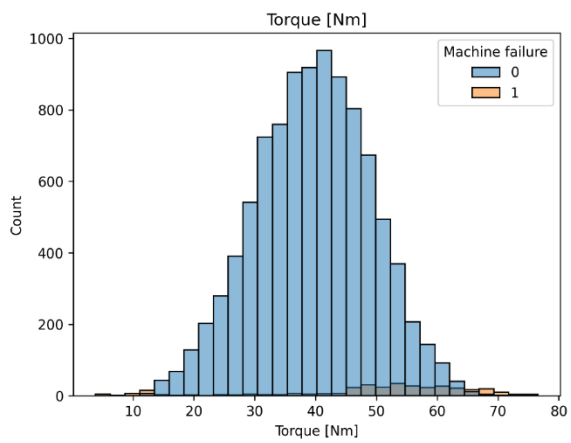


Figure 6.6 Rotational Speed [rpm] Distribution highlighting a right-skewed operational profile

6.5.3 Correlation Analysis

The correlation heatmap reveals important relationships between features:

- A strong positive correlation exists between Air Temperature and Process Temperature, indicating consistent environmental behavior.
- A strong negative correlation is observed between Rotational Speed and Torque, reflecting the mechanical trade-off in machine operation.

These correlations highlight that machine failure is influenced by multiple interacting variables rather than a single factor.

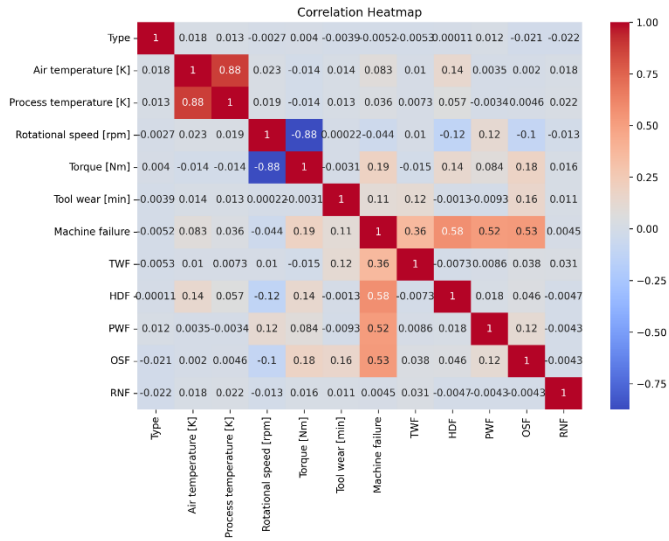


Figure 6.7 Correlation Heatmap illustrating the dependencies between industrial sensor variables.

6.5.4 Confusion Matrix Interpretation

The confusion matrices clearly show differences in model reliability:

- XGBoost minimized false negatives, ensuring that most actual failures were correctly detected, which is critical in industrial applications.
- Logistic Regression produced many false positives, which in real-world systems could lead to unnecessary maintenance actions and operational inefficiency.

This confirms that ensemble models provide more stable and reliable predictions for predictive maintenance tasks.

6.6 Discussion of Results

The results confirm that ensemble learning models, particularly XGBoost, are highly effective for predictive maintenance applications. The complexity of machine failure behavior, which depends on multiple interacting features such as torque, temperature, and tool wear, is better captured by non-linear models.

The application of SMOTE played a crucial role in addressing class imbalance. Without it, simpler models tended to favor the majority class, leading to poor failure detection performance. After balancing the dataset, model performance improved significantly, especially in recall and F1-score.

Overall, XGBoost demonstrated superior generalization ability due to its gradient boosting mechanism, which effectively learns complex patterns in the dataset.

6.7 Practical Interpretation

From a real-world perspective, the developed XGBoost model can be implemented as a predictive monitoring system in smart manufacturing environments.

By continuously monitoring key variables such as Torque, Rotational Speed, and Tool Wear, the system can detect early signs of potential failure. When abnormal patterns are identified, alerts can be triggered to allow maintenance teams to take proactive action before breakdowns occur.

This approach reduces unexpected downtime, improves operational efficiency, and supports cost-effective maintenance strategies.

6.8 Chapter Summary

This chapter evaluated and compared multiple machine learning models for predictive maintenance. The results clearly demonstrate that while traditional models struggle with imbalanced industrial data, ensemble methods, especially XGBoost provide highly accurate and reliable predictions.

With a strong ROC-AUC score and balanced precision-recall performance, XGBoost is identified as the most suitable model for machine failure prediction in this study.

Chapter 7: Conclusion and Future Work

7.1 Conclusion

This project developed a machine learning-based predictive maintenance system using the AI4I 2020 dataset to detect potential machine failures. Key sensor features such as temperature, rotational speed, torque, and tool wear were analyzed to demonstrate data-driven solutions in smart manufacturing.

The dataset showed class imbalance, with fewer failure cases, which was addressed using SMOTE for balanced training. Multiple models were implemented, including Logistic Regression, Decision Tree, Random Forest, XGBoost, SVM, and KNN for comparison.

Ensemble models performed best, with XGBoost achieving the highest F1-score and ROC-AUC, followed by Random Forest. Overall, the results confirm that machine learning can enable early failure detection, reduce downtime, and support proactive decision-making in manufacturing.

7.2 Key Findings

The study identified several important findings that contribute to the effectiveness of predictive maintenance systems. Machine failure prediction was strongly influenced by operational features such as torque and tool wear, indicating their importance in monitoring system health. The dataset imbalance significantly affected model performance, making resampling techniques like SMOTE essential for accurate predictions.

It was also observed that ensemble learning models consistently outperformed traditional algorithms, with XGBoost providing the most reliable and accurate results. Additionally, proper data preprocessing and feature engineering played a critical role in improving model performance.

These findings reinforce the importance of combining data quality, appropriate modeling techniques, and advanced algorithms in building effective predictive maintenance solutions.

7.3 Future Work

Although the project achieved strong predictive performance, several enhancements can be made to improve real-world applicability. One important direction is the integration of machine learning models with real-time IoT systems, enabling continuous monitoring and instant failure detection in industrial environments.

Further improvements ca be achieved by exploring deep learning techniques such as Artificial Neural Networks, Long Short-Term Memory networks, and Convolutional Neural Networks, which are particularly effective for handling complex and time-series data. Additionally, deploying the model as a web or cloud-based system using technologies such as Flask, Django, or cloud platforms like AWS and Azure would enhance accessibility and usability.

Future research can also focus on cost-based optimization by incorporating the economic impact of different failure types into the model. Moreover, applying explainable AI techniques such as SHAP or LIME would improve transparency and help engineers better understand model predictions. Another potential improvement is multi-sensor fusion, where additional data sources such as vibration, sound, and pressure are combined to increase prediction accuracy and robustness.

7.4 Final Remarks

In conclusion, this project demonstrates the effectiveness of machine learning in predictive maintenance applications within smart manufacturing. By leveraging historical sensor data, the system provides a reliable approach for detecting potential failures and improving operational efficiency.

The strong performance of ensemble models, particularly XGBoost, confirms their suitability for industrial classification tasks. With further advancements such as real-time deployment, deep learning integration, and enhanced interpretability, this system can be extended into a fully functional intelligent maintenance solution capable of supporting modern industrial needs.